

DAE YON HWANG

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EDUCATION

University of Toronto	Ph.D. in Electrical & Computer Engineering, 4.0/4.0	Nov 2022
Texas A&M University	Master of Science in Electrical Engineering, 4.0/4.0	May 2016
Hanyang University	B.S. in Electronic Engineering, 3.56/4.0 (<i>Cum Laude</i>)	Feb 2014

WORK EXPERIENCE

Layer 6 AI @TD - Senior Research Machine Learning Scientist *Trustworthy AI* May 2025 - Present

- **Black-Box Agent Optimization**
 - Developed a **Sequential Monte Carlo framework** for test-time policy optimization in black-box LLM agents, bridging reinforcement learning and Bayesian inference
 - Trained a **lightweight critic model** to guide black-box LLM policies, **reducing optimization cost by 20%** while **improving performance by 16%** over the GRPO baseline across agentic tasks
 - Published research findings in **ICML 2026**
- **Conversational RAG with Safety Guardrails**
 - Deployed a production RAG system balancing model capability with strict banking compliance requirements
 - Designed query classification, rewriting, and abstention mechanisms to enforce policy-aligned responses
 - Implemented a **three-layer safety framework** and led **automated agentic red-teaming** to mitigate adversarial inputs and jailbreak attempts
 - Launched a public customer-facing chatbot on TD Insurance, maintaining **zero jailbreak incidents** in production
 - Formulated multi-turn safety guardrails with **conformal prediction guarantees** under dynamic risk bounds, with findings submitted to **ICLR 2027**
- **Vision-Language Model Interpretability**
 - Built a synthetic pipeline for **counterfactual chart generation** to systematically evaluate chart understanding in VLMs
 - Uncovered **memorization** and **shortcut learning behaviors** using both **heuristic benchmarking** and **mechanistic interpretability**
 - Published research findings in **EMNLP 2026**
- **Multi-Objectives Preference Optimization**
 - Formulated data topology based on **partially ordered sets** to model trade-offs in multi-objective preference alignment (e.g., helpfulness vs. harmlessness)
 - Evaluated across **DPO** and **listwise DPO**, achieving significant gains in alignment stability across up to **4 distinct objectives**
 - Under review at **ACL Rolling Review (ARR)** for upcoming ACL cycles

Amazon Science - Applied Scientist II *Artificial General Intelligence*

Sep 2022 - Apr 2025

- **Multimodal Foundation Model Post-Training**
 - Generated **synthetic multi-step reasoning data** for post-training multimodal foundation models
 - Developed **LLM-as-a-Judge pipelines** for data filtering and quality control in post-training
 - Designed and scaled **fine-tuning** and **preference-optimization methods** (e.g., SFT, DPO, PPO, GRPO) across multi-GPU clusters using **TRL** and **FSDP**
 - Contributed to post-training **Amazon Nova foundation models**, improving performance across general and domain-specific tasks including legal and finance
 - Named contributor on the **Amazon Nova Technical Report and Model Card**

- **Foundation-Model Recommendation System**

- Developed and deployed a foundation-model recommendation system combining **instruction tuning, preference optimization**, and **RAG**
- Designed online evaluation and continuous-feedback mechanisms and validated the system through customer-facing experiments
- Deployed the system on Prime Video, achieving a **20% improvement in positive-feedback rate** over the control model in online experiments

- **Information Retrieval for Alexa**

- Developed **LLM-based synthetic-data generation** and **representation-learning methods** to improve zero-shot retrieval across diverse scenarios
- Designed **dimensionality-reduction methods** that **accelerated retrieval efficiency by 5×** while maintaining performance
- Published **3 papers** in top-tier venues, including **ACL** and **EMNLP**

University of Toronto - Research Assistant *Biometrics Security Lab*

Sep 2018 - Sep 2022

- **Robust User Verification from Physiological Signals**

- Developed **multimodal physiological-signal models** for robust user verification and adversarial resilience
- Collected and released physiological signals from **170 participants**, creating a large public dataset for biometric research
- Published **5 papers** in top-tier venues, including **IEEE TIFS** and **ICASSP**

- **Human Activity Recognition with Wearable Device**

- Developed **hierarchical multimodal models** combining inertial and physiological sensors for wearable activity recognition
- Published research findings in **ICASSP**

SELECTED PUBLICATIONS

D. Y. Hwang , et al., "Agentic Monte Carlo: Simulating Reinforcement Learning for Black-Box Agents"	ICML 2026
Y. Jiang, D. Y. Hwang , et al., "Chartographer: Counterfactual Chart Generation for Evaluating Vision-Language Models"	EMNLP 2026
D. Y. Hwang (<i>Contributor</i>), "The Amazon Nova Family of Models: Technical Report and Model Card"	Amazon Science 2024
D. Y. Hwang , et al., "Link, Synthesize, Retrieve: Universal Document Linking for Zero-Shot Information Retrieval"	EMNLP 2024
D. Y. Hwang , et al., "EmbedTextNet: Dimension Reduction with Weighted Reconstruction and Correlation Losses for Efficient Text Embedding"	ACL 2023
D. Y. Hwang , et al., "GAN-LM: Generative Adversarial Network using Language Models for Downstream Applications"	INLG 2023
D. Y. Hwang , et al., "PBGAN: Learning PPG Representations from GAN for Time-Stable and Unique Verification System"	IEEE TIFS 2021
D. Y. Hwang , et al., "Evaluation of the Time Stability and Uniqueness in PPG based Biometric System"	IEEE TIFS 2020

SELECTED RESEARCH TALKS

From Generative AI to Agentic AI: Building Trustworthy AI	Kyungpook National University
Introduction to Mechanistic Interpretability	TD Bank
Integrating GANs with Language Models	Amazon

PROFESSIONAL SERVICE

Reviewer: EMNLP, ACL, ARR, NeurIPS, IEEE JBHI, IEEE TIFS

Program Committee: EMNLP 2023 Industry Track

SKILLS

Research: Foundation Models, LLM Post-Training, Reinforcement Learning, Agentic AI, Reasoning, Multimodal Learning, Trustworthy AI, Mechanistic Interpretability, Safety & Guardrails

Engineering: Python, PyTorch, C++, vLLM, Ray, Hugging Face (TRL, PEFT, Accelerate), FSDP, DDP, AWS, CUDA

Languages: English (Native/Fluent), Korean (Native)